

Jack mackerel management strategy evaluation for SC14

Operating models, candidate management procedures, robustness tests, and
matters for Scientific Committee consideration

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Jack Mackerel Management Strategy Evaluation: Synthesis for SC14

Status and purpose. This report synthesizes the June 2026 SCW17 MSE workshop and subsequent model and management-procedure development.

Executive summary

The jack mackerel MSE has progressed from a broad workshop design into a narrower comparison of candidate management procedures (CMPs) tested against a reference operating model (OM) and a structured set of robustness OMs. SCW17 established the principal design choices: simple and interpretable empirical rules, simulation of correlated index errors, dynamic reference points, explicit robustness testing, a broad performance-measure set, and development of an exceptional-circumstances protocol. Post-workshop work corrected and extended the implementation, tuned the leading rule families, and examined their performance under one-stock and two-stock hypotheses, recruitment stress, movement, selectivity, and alternative annual TAC-change limits.

A novel feature of this MSE is its treatment of the five abundance indicators as related observations rather than independent pieces of evidence. The OMs simulate their observation errors jointly using historical estimates of each series' uncertainty, temporal autocorrelation, and cross-correlation. This allows the evaluation to represent shared movements among indicators, including correlation that may arise when the population shifts spatially. The estimated correlation structure used for the reference OM is illustrated in Figure 1.

Among the dozens of MPs evaluated at the SCW17 workshop, the analysts concluded that the hockey-stick and power-ramp designs represented by these two CMPs, together with reasonable variants of those designs, were suitable to carry forward as proposals for consideration at SC14.

This analytical conclusion identifies a focused set for Scientific Committee consideration; it is not a formal recommendation or adoption decision.

The [main Jack Mackerel MSE resource page](#) describes the naming conventions and how the MSE components fit together, and provides the CMP and OM registries and translation tables linking the labels used across analyses and reports.

Two CMPs are the focal cases for the current comparison:

- **HS-20 (MP43)** uses a hockey-stick harvest control rule (HCR), a 2,000-kt catch target, a 270-kt minimum, and annual limits of a 20% decrease and 15% increase.
- **PR-20 (MP44)** uses a power-ramp HCR, a 1,500-kt catch target, the same minimum, and the same annual limits, with a more responsive reduction in advice as the indicator declines.

Both use the combined indicator from five standardized abundance indices. The earlier **HS+20 (MP29)** and **PR+20 (MP32)** specifications are retained as sensitivity cases. They arose from an initial request informed by early feedback from Chile’s internal MSE research project; their 20% maximum annual TAC increase was not selected from the later Commission poll. That poll helped inform the broader set of annual TAC-change variants examined here. The remaining **HSsym**, **HS-30**, **PRsym**, and **PR-30** procedures are additional comparison cases. The paired catch trajectories for focal HS-20 and sensitivity HS+20 illustrate one annual-limit contrast in Figure 4.

This distinction matters for interpreting the illustrative scorecard. With all eight CMPs now tuned on the same 500 draws, HS+20 ranks first under both shown weighting schemes, while PR+20 ranks seventh; the six variants occupy positions throughout the ordering. That order is a relative trade-off result, not an acceptability test. It depends on the selected metrics, directions, scaling, and weights, and should be read alongside the raw outcomes and uncertainty. The scorecard ordering does not determine which cases are focal: focal status records the present analytical emphasis, whereas the scorecard summarizes relative trade-offs under selected metrics and weights.

All eight CMPs were refined against the same 500 posterior draws to target an approximately 60% mean probability of dynamic Kobe-green status during 2041–2050. Their realized probabilities range from 59.2% to 61.0%, so the comparison is now like-for-like with respect to the tuning objective. The archived 100-draw results remain useful provenance but are superseded by the 500-draw reference and robustness results used throughout this report. The earlier comparisons reviewed on 24 July identified a broad contrast between the stability of the hockey-stick family and the stronger response of the power-ramp family as indicators declined. The completed results refine that comparison across all eight CMPs; differences depend on the OM and performance measure and do not establish a preferred or “optimal” CMP.

The most consequential robustness finding came from the two-stock hypotheses. Without movement, the southern component passed the agreed benchmark while the northern component required an exploratory change of ~3% in simulated catch distribution from north to south

to reach the same benchmark. This is a sensitivity result, not an allocation recommendation. With movement, no tested fixed catch distribution achieved the long-term benchmark for both components. Equal weighting of indices could also mask a declining southern component when northern indices increased. Separate annual tracking of northern and southern indices is therefore an important candidate element of exceptional-circumstances monitoring.

The Scientific Committee is asked to consider the adequacy of HS-20 (MP43) and PR-20 (MP44) as the focal cases; the role of HS+20 (MP29) and PR+20 (MP32) as sensitivity cases; the other annual TAC-change variants to retain; the role of all five indices, including the only fishery-independent series; the interpretation of two-stock robustness; the completed 500-draw reference and robustness evaluation; and the work needed to validate and complete banking-and-borrowing and exceptional-circumstances evaluation.

Purpose, scope, and decision status

The MSE is intended to compare complete management procedures in a closed-loop simulation before one is considered for implementation. Each CMP converts agreed observations into annual catch advice; each OM represents a plausible description of stock and fishery dynamics against which the CMP is tested. Performance measures describe biological outcomes, yield, stability, and fishery feasibility over specified periods.

This report melds the technical foundation assembled in the [benchmark-preparation and MSE-planning materials](#), the design record from the [SCW17 MSE Workshop report](#), and the post-workshop analysis summarized in the [final 24 July task-team meeting report](#) and the [presentation from Wageningen Marine Research consultants \(WUR\)](#). The preparation site preceded SCW17 and provided the detailed model specifications, diagnostics, working papers, and technical discussions needed to update the operating models used at SCW17 and in subsequent MSE work. Their different roles are summarized in Table 1.

Table 1: Source materials and their roles in this synthesis.

Source	Role in this synthesis
Benchmark preparation and MSE planning	Pre-SCW17 benchmark papers, OM specifications, diagnostics, index analyses, and preparatory meeting records that supported updates to the OMs used at SCW17 and beyond.
SCW17 workshop report, 15–19 June 2026	Agreed MSE design, OM and CMP principles, reference-point approach, performance framework, and planned robustness work.
Post-workshop jmMSE analysis	Implemented CMP definitions, tuning runs, reference and robustness simulation outputs, and reproducibility records.

Source	Role in this synthesis
Task-team meeting, 24 July 2026	Interpretation of results, wording cautions, issues for SC14, and agreed follow-up.
July WUR presentation	Detailed tuning sequence, OM contrasts, and graphical robustness diagnostics available at the time of review.
Draft weighted scorecard, 23 July 2026	Example of an editable multi-criteria comparison method; not an exhaustive candidate list, metric list, or decision.

The Scientific Committee provides scientific advice and the Commission weighs management trade-offs and makes management decisions. A shortlist in an MSE analysis is not an adoption decision.

Development from SCW17 to the current analysis

SCW17 continued work begun at the [2025 SCW15 MSE workshop](#) (SPRFMO Jack Mackerel Working Group 2025) and the 2026 benchmark. The [benchmark-preparation and MSE-planning site](#) preserves the detailed model specifications, diagnostics, index working papers, and preparatory meeting discussions that preceded SCW17. That record provided the technical basis for updating the operating models taken into the workshop and used in subsequent MSE analyses.

SCW17 aimed to prepare an OM set, trial CMPs, performance statistics, robustness scenarios, and documentation for SC14. The workshop favored simple empirical rules because they are transparent, reproducible, and can be locked before use. Trial CMPs were used to learn about rule behavior and were not final recommendations.

Post-workshop work made the following material advances:

1. Index projection errors were extended to include iteration-specific variability, autocorrelation, and cross-correlation among index deviations.
2. Dynamic biological reference points and additional measures, including vulnerable-biomass and exploitation measures, were incorporated.
3. A straight-slope comparison rule was added.
4. The power-ramp implementation was corrected to remove discontinuities around its limit. The correction was required for consistent implementation but did not overturn the broad workshop conclusions.
5. A projected effort constraint was added to avoid implausible catches that would require extreme effort.
6. CMP tuning was performed as a nested screen of HCR family, catch target, minimum catch, index limit, index set, simulation replication, and robustness performance.

7. Alternative annual TAC-change limits were retuned instead of being compared using triggers tuned under a different constraint.

The development of vulnerable-biomass performance indicators responded to a request from SCW17 for a measure that could complement spawning biomass and be more relatable to fishers. In this context, vulnerable biomass represents the fish available to the fishing fleet under the modeled selectivity. It was therefore developed as a fishery-facing performance measure, while spawning biomass remains the principal measure of reproductive stock condition.

The July WUR presentation used the word “optimal” for some within-family runs. The 24 July meeting clarified the interpretation: those runs were useful within-family representatives, but the analysis does not establish an optimal procedure across management objectives.

Operating models and simulation methods

Reference and robustness set

The current grid centers on the one-stock benchmark model `h1_0.16` with steepness 0.65. Robustness OMs test major structural and parametric uncertainties rather than assigning probability to one forecast (Table 2).

Table 2: Reference and robustness operating-model set.

OM label	Main contrast	Analytical role
om11	One stock, model 0.16, steepness 0.65	Reference OM for current tuning
om11_1	Alternative selectivity	Younger/smaller fish exposed to greater fishing pressure
om11_2	One-year recruitment crash	Acute recruitment stress
om11_3	Seven-year cyclic recruitment reduction	Prolonged recruitment stress
om12	One stock, steepness 0.8	Productivity robustness
om13	One stock, SC13-era/model 1.14 configuration	Assessment/configuration robustness
om21	Two stocks, no movement	Stock-structure and catch-distribution robustness
om21_1	Two stocks with movement	Spatial redistribution and indicator-conflict robustness
om22	Two stocks, steepness 0.8	Two-stock productivity robustness

OM label	Main contrast	Analytical role
om23	Two stocks, model 1.14 configuration	Two-stock assessment/configuration robustness

The OM set (reference and robustness) is not a declaration that all states are equally plausible. Its purpose is to expose CMP sensitivity to assumptions that could matter to management outcomes. Any later OM weighting should be explicit and documented.

Recruitment, observation, and implementation uncertainty

Recruitment variability strongly influences stock trajectories and dynamic reference points. The core simulations therefore retain recruitment uncertainty, while discrete crash and cyclical-reduction cases act as stress tests. SCW17 treated ENSO and climate effects as robustness questions at this stage rather than embedding a single climate forecast in the core OM.

Dynamic reference points

Environmental recruitment variability can make static reference points misleading. In [paragraph 58 of the SCW17 workshop report](#), the Working Group strongly recommended calculating the ratio B_{MSY}/B_0 within each MCMC draw, multiplying it by the corresponding dynamic B_0 , and evaluating spawning biomass relative to that dynamic B_{MSY} . The current tuning objective uses a dynamic Kobe-green measure that combines biomass and fishing-pressure status.

Reference-point choices remain consequential. In particular, alternative selectivity can change F_{MSY} and therefore F/F_{MSY} even when biomass and catch trajectories are similar. Interpretation should separate changes caused by reference-point definitions from changes in underlying CMP behavior.

Generating OM data indices

Each CMP receives simulated abundance indices rather than true stock status. Projected index deviations include autocorrelation and cross-correlation so that the observation model better represents the historical behavior of multiple surveys and CPUE series. Simulation diagnostics should show the distributions of index-error standard deviations, autocorrelation parameters, and correlations used.

The combined indicator does not assume that five series provide five independent confirmations of stock change. In each OM iteration, projected index errors are generated jointly from the historical index deviations: series-specific log-scale standard deviations describe uncertainty,

lag-1 parameters preserve temporal autocorrelation, and an iteration-specific correlation matrix preserves cross-correlation among indices. These relationships matter because common environmental effects, fleet behavior, and spatial redistribution of the population can cause several indices to rise or fall together.

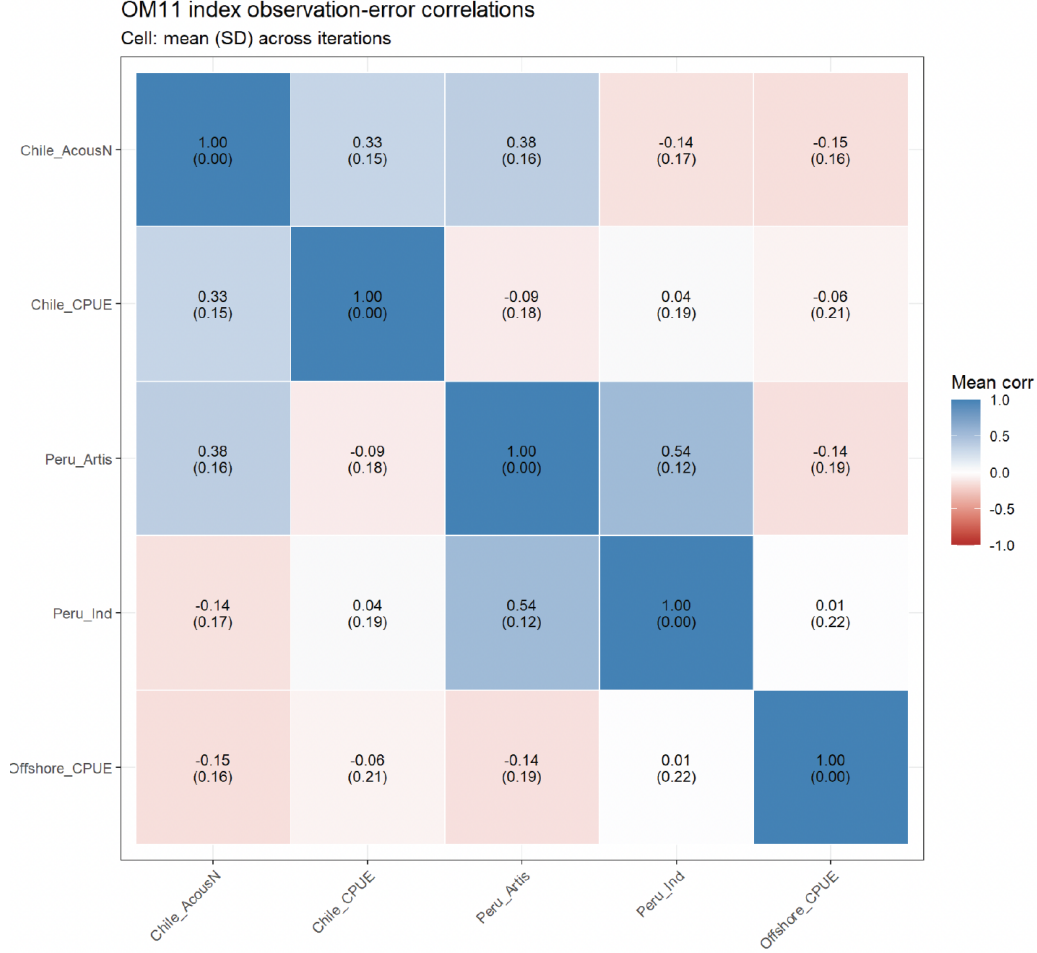


Figure 1: OM11 index observation-error correlations. Each cell gives the mean correlation across posterior iterations, with its between-iteration standard deviation in parentheses. Positive values indicate index errors that tend to move together; negative values indicate opposing movements.

As shown in Figure 1, the estimated relationships are neither uniformly positive nor interchangeable. For example, the Peru artisanal and industrial indices have the strongest positive mean cross-correlation (0.54), whereas several relationships involving the offshore CPUE series are weakly negative. The standard deviations in parentheses show that the observation-error correlation matrix also varies among OM iterations.

The resulting closed-loop test therefore exposes the CMP to realistic combinations of agreement, redundancy, and divergence among indicators. The novelty lies in evaluating the simple, transparent equal-weight composite under this estimated multivariate observation process; equal weighting itself is not covariance weighting and does not mathematically remove correlation. This approach is consistent with the MSE best-practice principle that management procedures should be tested using simulated management data that represent observation uncertainty rather than receiving true stock status (Punt et al. 2016).

Candidate management procedures

Common indicator and annual calculation

The central CMPs use five observed series: northern Chile acoustic biomass, Chilean CPUE, Peruvian artisanal and industrial indices, and offshore CPUE. Each series is expressed relative to its 2019–2023 mean. The standardized values are combined with equal weights within each year, after the OMs have generated the indices using estimated uncertainty, temporal autocorrelation, and cross-correlation among series. The CMP then uses a three-year recent average of the combined indicator.

Most data are assumed to arrive with a one-year lag, while offshore CPUE has a two-year lag. Under the tested timing, a calculation during 2026 for 2027 advice would use 2025 values for the acoustic, Chilean CPUE, and Peruvian indices, and the 2024 offshore CPUE value. Same-year 2026 observations do not enter that tested 2027 calculation unless the timing rule is explicitly changed and re-evaluated.

Naming convention and candidate variants

The working labels encode the HCR family and the most distinguishing annual TAC-change constraint. HS means hockey-stick and PR means power ramp. +20 identifies the base 20% maximum increase, -20 identifies the reverse asymmetry with a 20% maximum decrease, sym identifies symmetric 15% limits, and -30 identifies a 30% maximum decrease. These labels should be used in figures and interpretive prose. The legacy MP identifiers remain in the table only to preserve the link to code, saved outputs, and earlier documents. The complete mapping is given in Table 3.

Focal cases and annual-limit sensitivities

All eight CMPs use the same five-index mean, 2019–2023 reference period, three-year averaging window, 270-kt minimum catch, 0.1 index limit, and 1,385-kt initial simulated catch. Each variant was refined under the reference OM using the same 500 posterior draws after setting its annual constraint.

The nested development framework in Figure 2 shows how the focused comparison was constructed: first select the HCR shape, then its target catch, and finally tune the trigger controlling the indicator-to-catch response, while retaining the common minimum catch and index limit. The figure records the earlier 100-draw trigger controls used during development; the subsequent common 500-draw refinement used for current results is reported in Table 3.

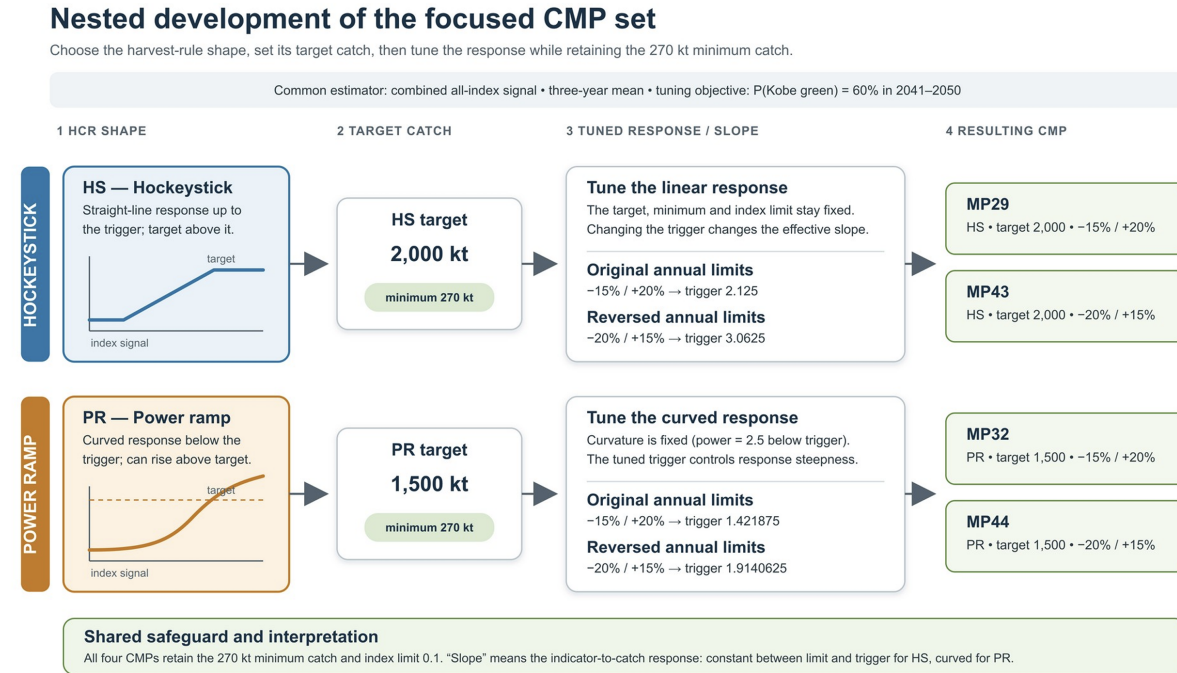


Figure 2: Nested development of the focused CMP set from HCR family through target catch and tuned response to the resulting HS and PR cases. Trigger values in this development diagram are the earlier 100-draw controls; the current 500-draw refined controls are given in Table 3.

The matched example in Figure 3 isolates the effect of HCR shape by assigning both rules the same minimum, target, limit, and trigger. The hockey-stick increases catch advice linearly and then remains at its target. The power ramp is more conservative below the trigger but continues above the target when the index exceeds the trigger. Its second panel illustrates the harvest-rate pattern that would follow **if** vulnerable biomass were directly proportional to the index. For both HCR forms, the implied harvest rate rises at low stock sizes because catch advice is not allowed to fall below the 270-kt minimum; that fixed catch floor becomes an increasing proportion of the vulnerable biomass as stock size declines. The panel is explanatory rather than an estimated performance result or an additional rule used by the CMPs.

Hockeystick and power-ramp responses to the same index

Matched example to isolate shape: minimum catch 270 kt; target 2,000 kt; index limit 0.1; trigger 2.

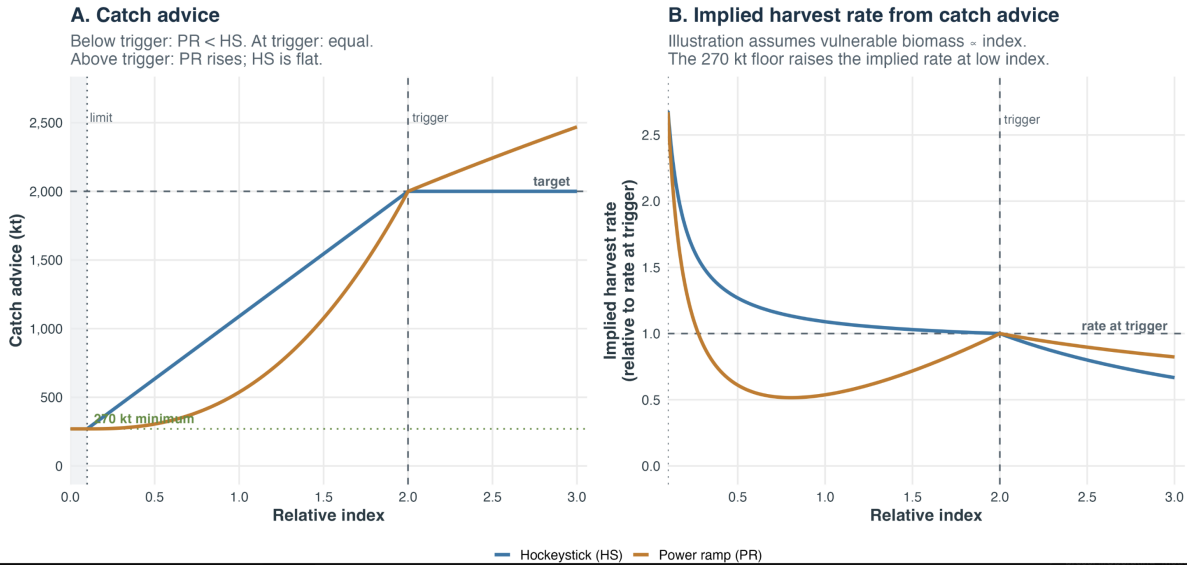


Figure 3: Matched illustration of hockey-stick and power-ramp responses to the same relative index. Both examples use a 270-kt minimum catch, 2,000-kt target, 0.1 index limit, and trigger of 2 solely to isolate rule shape; the focal HS and PR CMPs retain their own targets and refined triggers listed in Table 3. The implied-harvest-rate panel assumes vulnerable biomass is proportional to the index and is conceptual rather than a simulated performance result.

Table 3: Naming convention, legacy identifiers, and definitions of the eight candidate variants.

Working label	Legacy ID	HCR family	Target or breakpoint (kt)	Tuned trigger	Annual decrease/increase limits
HS+20	MP29	Hockey-stick	2,000	2.125	-15% / +20%
HS-20	MP43	Hockey-stick	2,000	1.9375	-20% / +15%
HSsym	MP45	Hockey-stick	2,000	2.0625	-15% / +15%
HS-30	MP47	Hockey-stick	2,000	2.0000	-30% / +20%
PR+20	MP32	Power ramp	1,500	1.640625	-15% / +20%
PR-20	MP44	Power ramp	1,500	1.6015625	-20% / +15%
PRsym	MP46	Power ramp	1,500	1.6250	-15% / +15%
PR-30	MP48	Power ramp	1,500	1.578125	-30% / +20%

The 500-draw refinement began from the earlier 100-draw tuned controls rather than restarting from a broad search. It then evaluated nearby trigger values under one frozen OM and common random-number configuration. The resulting dynamic Kobe-green probabilities for HS+20, HS-20, HSsym, HS-30, PR+20, PR-20, PRsym, and PR-30 are 61.0%, 59.2%, 60.4%, 60.4%, 60.3%, 59.8%, 59.5%, and 60.6%, respectively.

A completed local trigger check tested one grid step below and above every selected control using the same frozen 500 draws. The steps were 0.0625 for the hockey-stick rules and 0.03125 for the power-ramp rules, matching the final bisection resolution for each family. All eight checks behaved as expected: a higher trigger produced a higher long-term Kobe-green probability and a lower trigger produced a lower probability. Across the eight CMPs, one lower step changed the probability by -1.36 to -1.74 percentage points and one upper step changed it by +1.10 to +1.84 points. In plain language, this is a local tolerance profile: it checks how much the tuning result changes after a small move on either side of the chosen trigger, rather than assuming the selected number is uniquely precise. The full values are saved in the [trigger-tolerance summary](#).

The hockey-stick increases advice linearly between the limit and trigger and then reaches a plateau. The power ramp reduces catch more quickly when the indicator is below its trigger and can continue to increase above its stated breakpoint according to the implemented rule. These structural differences, not the labels alone, create the main responsiveness-versus-stability contrast.

The completed 500-draw refinement is a higher-replication evaluation of the same conceptual CMPs, not a new set of procedures. Its controls and performance outputs are now the common basis for the candidate summaries in this report.

Annual TAC-change variants

The base CMPs limit annual advice to a 15% decrease and a 20% increase. The task team identified a limited set of alternatives for retuning and comparison, summarized in Table 4.

Table 4: Annual TAC-change-limit variants by HCR family.

HCR family	Base	Reverse asymmetry	Symmetric	Larger decrease
Hockey-stick	HS+20: -15%/+20%	HS-20: -20%/+15%	HSsym: -15%/+15%	HS-30: -30%/+20%
Power ramp	PR+20: -15%/+20%	PR-20: -20%/+15%	PRsym: -15%/+15%	PR-30: -30%/+20%

The variants change both management responsiveness and the tuned trigger. They should therefore be compared as complete, retuned CMPs. The available 500-draw reference and nine-OM robustness runs support the common higher-replication comparison reported here.

The following reference-OM plot shows 15 reproducibly selected simulation iterations for the two hockey-stick asymmetry cases over 2025–2050. Iteration IDs are paired exactly across panels: for example, iteration 6 would be shown for both MPs if selected, and every selected ID occurs once in each panel. Each iteration has its own line color, and the same iteration uses the same color in both panels. HS-20 is MP43 (**tun43**), with annual limits of -20%/+15%; HS+20 is MP29 (**tun29**), with limits of -15%/+20%. The vertically stacked panels share a common catch scale, as shown in Figure 4.

Choice of indices

A CPUE/fishery-index-only comparison was more responsive but produced slightly lower short-term catch and higher short-term catch variability. The task team retained all five indices for the current focal comparison. The northern Chile acoustic survey is the only fishery-independent abundance index in the set and provides information that fishery-dependent series cannot fully replace. Conversely, equal weighting across regions can become problematic if northern and southern components diverge. These two findings support retaining the acoustic series while also monitoring regional index groups separately.

Projected catch trajectories: reference operating model

The same 15 reproducibly selected iterations and matching colors are shown for each MP, 2025-2050

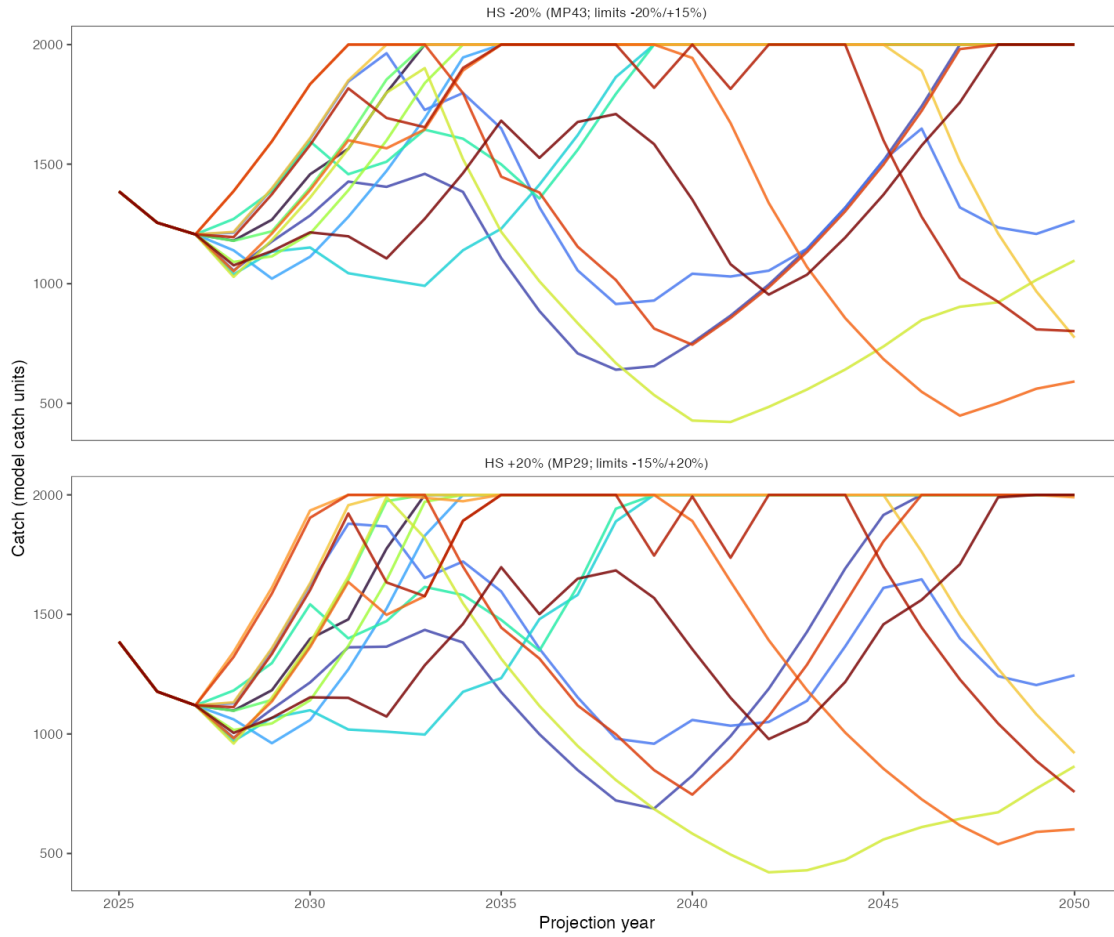


Figure 4: Projected catch trajectories for the same 15 simulation iteration IDs under both hockey-stick MPs in the reference operating model. Each iteration retains the same line color in both panels so matching trajectories can be traced through time and across MPs.

Performance evaluation and the draft scorecard

Performance framework

SCW17 agreed that CMPs should be evaluated across safety, yield, stability, and feasibility rather than by the tuning target alone. Tuning several CMPs to the same Kobe-green objective makes that objective a control for comparison, not a sufficient ranking statistic. The full performance framework can include:

- probabilities of dynamic Kobe-green, low biomass, overfishing, and stock collapse;
- spawning biomass, vulnerable biomass, and fishing pressure relative to reference points;
- short-, medium-, and long-term catch;
- interannual catch change and the frequency with which TAC-change limits bind;
- low-catch, minimum-catch, and fishery-shutdown-like outcomes;
- effort or catchability feasibility diagnostics; and
- robustness measures across individual OMs and stress tests.

A short headline table can aid Scientific Committee review, but the complete metric set and underlying distributions should remain available in the technical annex and interactive outputs.

The augmented candidate exports include vulnerable biomass relative to its 2025 level and relative to equilibrium vulnerable biomass at the fishing mortality that produces MSY. Vulnerable biomass was selected to provide a performance measure related to the fish available to the fishing fleet under the modeled selectivity, complementing spawning-biomass measures of biological stock condition. Reference-OM results are shown in Figure 5 and the corresponding robustness results in Figure 6.

Performance metrics by CMPs

The quilt in Figure 7 is an unweighted comparison of the eight annual-change variants listed in Table 3 under the reference OM. Annual values were averaged over 2041–2050 within each simulation iteration; cell labels are medians of those iteration means. Color is normalized independently within each metric across the eight variants.

Purple means lower relative preference and light lavender means higher relative preference (0 is darkest; 1 is lightest). Higher SB/SB_{MSY} , catch, VB/VB_{2025} , and VB/VB_{MSY} , and dynamic Kobe-green probability are treated as better; lower F/F_{MSY} , IACC, dynamic Kobe-red probability, probability of catch below 270 kt, and mean annual catch reduction are treated as better. Lower F/F_{MSY} represents lower fishing pressure, while catch is shown separately. The within-column color scale is not an absolute biological or management threshold and must not be interpreted as a rank or recommendation.

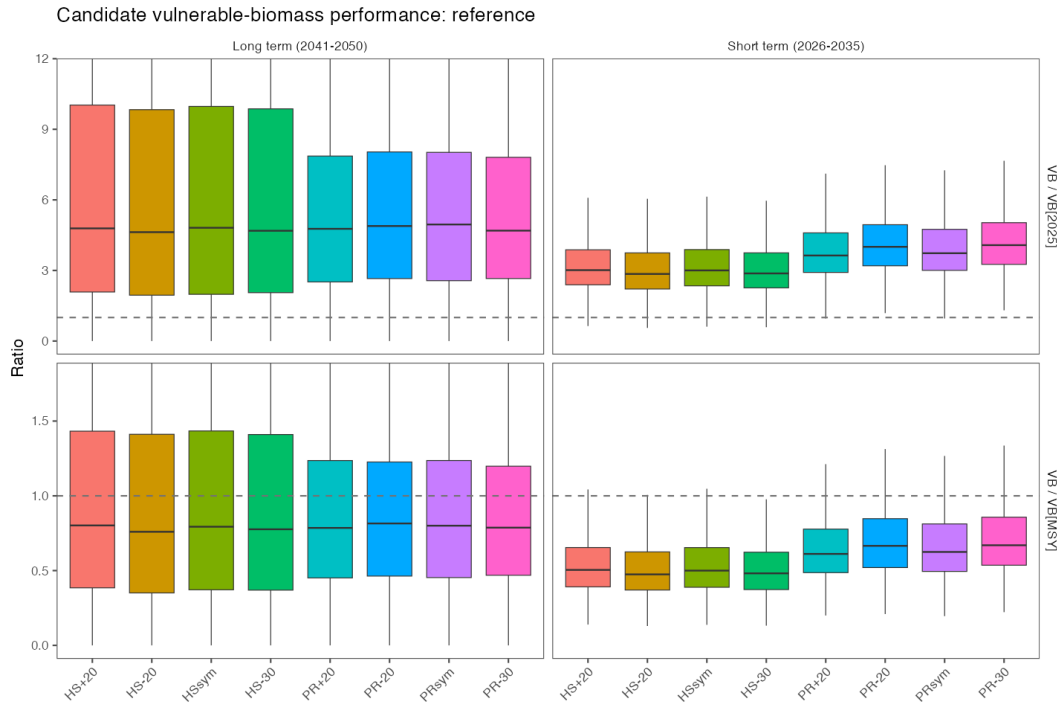


Figure 5: Vulnerable-biomass performance for the reference operating model.

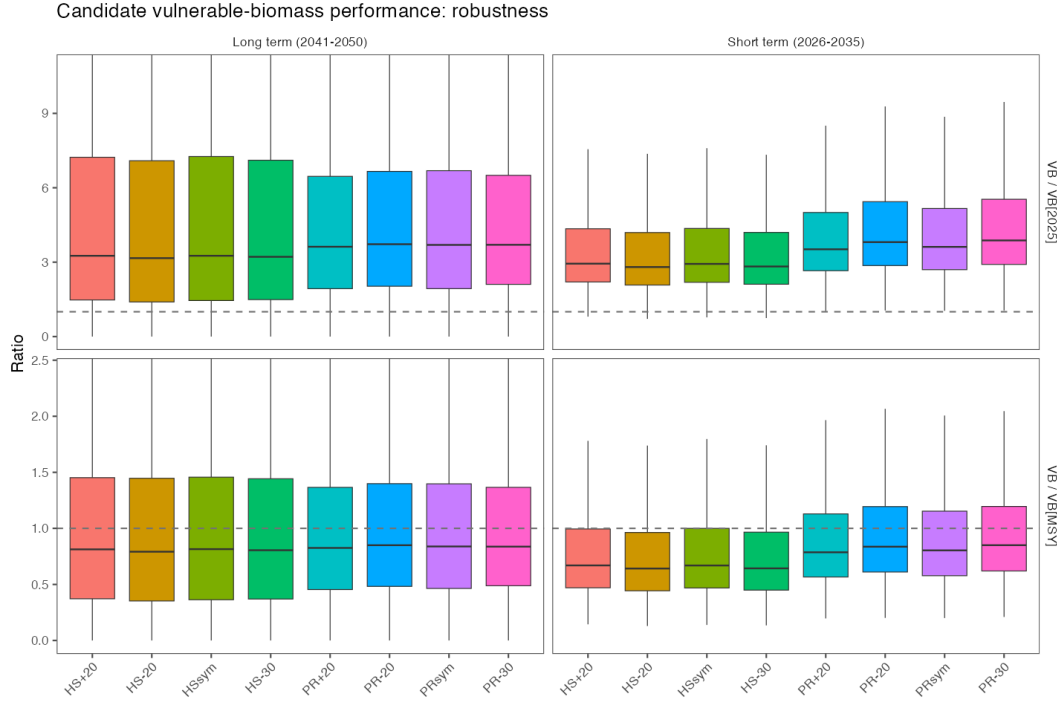


Figure 6: Vulnerable-biomass performance across the five single-stock CJM robustness operating models. The four two-stock operating models are omitted from this pooled figure because their Northern and Southern components require stock-specific interpretation and are examined separately in the robustness results.

The Kobe probabilities use the same dynamic biomass denominator as the tuning objective: the equilibrium SB_{MSY}/SB_0 fraction is applied to projected unfished spawning biomass in each year and iteration. The low-catch probability uses 2041–2050; mean catch reduction averages positive annual percentage reductions during 2026–2050. These four diagnostics are displayed in the quilt but are not automatically added to the illustrative composite score below, avoiding an undocumented change in weights and partial double-counting of stock status, fishing pressure, catch, and IACC.

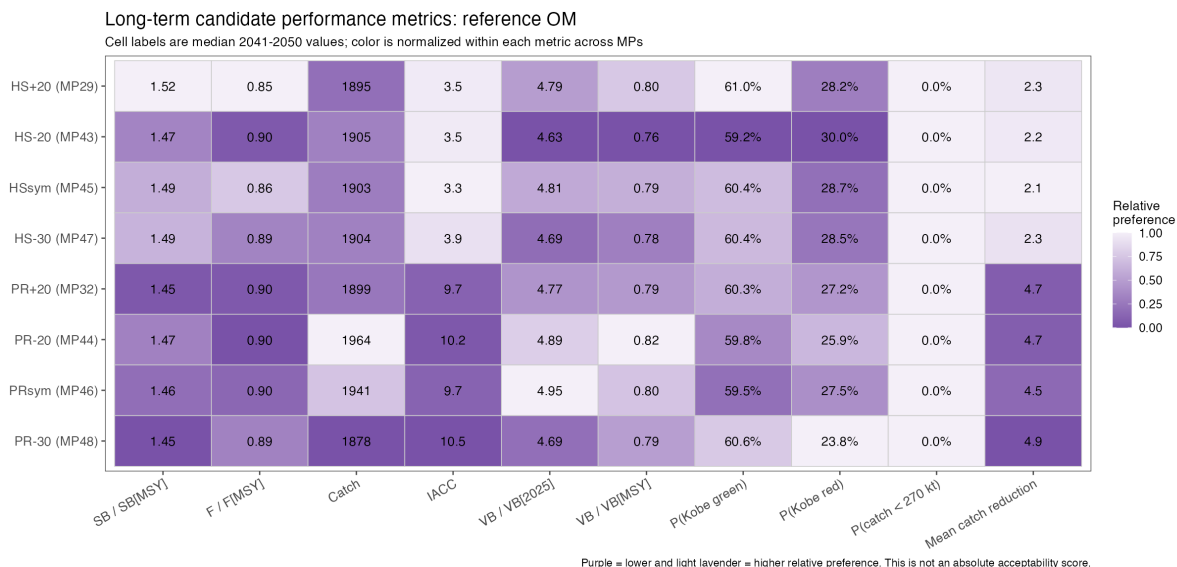


Figure 7: Reference-OM candidate performance metrics for the eight naming-convention variants. Cell labels are median iteration summaries, except the dynamic Kobe probabilities, which are means over all iteration-years. Purple indicates lower relative preference and light lavender indicates higher relative preference within each metric (0 is darkest; 1 is lightest).

The underlying values, preferred directions, metric ranges, and normalized scores are saved in [candidate_quilt_reference_summary.csv](#). An editable scorecard input layer is saved in [candidate_scorecard_input_reference.csv](#). It converts quilt relative preference to a 0–100 benefit score, records the preferred direction and raw value, and supplies editable **include** and **weight** columns. The default weight is one for every metric, but no weighted total is calculated in this export. Changing weights or aggregating scores remains an explicit user decision under the method below.

Extensible weighted-scorecard method

The scorecard should be read as three connected layers, with no layer replacing the one before it:

1. **Raw performance and uncertainty.** Report the performance statistic in its original units together with its simulated distribution or an appropriate interval. This is the primary evidence layer and includes, for example, catch in kt, biomass and fishing-pressure ratios, probabilities, and IACC. The boxplots and uncertainty bars elsewhere in this report provide examples. Point estimates alone are insufficient where CMP distributions overlap.
2. **Threshold- or target-based absolute performance.** Where the Scientific Committee or Commission has agreed a biologically or operationally meaningful target and limiting boundary, performance can be expressed relative to those fixed reference values. Potential applications include dynamic Kobe probabilities, biomass thresholds, low-catch risk, and an agreed tolerance for interannual catch variability. The target, limiting boundary, direction, time window, and treatment of uncertainty must be specified before calculating this layer; they must not be inferred from the candidate results.
3. **Range-normalized relative preference.** For metrics without agreed absolute targets, the quilt can scale each CMP between the least and most preferred observed values. This supports comparison across unlike units but describes position within the current candidate set, not absolute performance or acceptability.

For an absolute higher-is-better metric with agreed limiting boundary L_m and target $T_m > L_m$, one possible transparent score is

$$a_{jm} = \min\left(1, \max\left(0, \frac{x_{jm} - L_m}{T_m - L_m}\right)\right).$$

For a lower-is-better metric with $T_m < L_m$, the numerator and denominator are reversed: $a_{jm} = \min(1, \max(0, (L_m - x_{jm})/(L_m - T_m)))$. These forms are illustrative; an agreed probability, risk tolerance, or other metric-specific function may be preferable. Unlike min-max normalization, fixed L_m and T_m values permit comparison across candidate sets and report versions.

The current worked scorecard implements **Layer 3**, using the performance quilt in Figure 7 as a reproducible relative trade-off calculation for the eight candidate variants under the reference OM. It does not yet calculate Layer 2 absolute scores because the necessary metric-specific targets and limiting boundaries have not been agreed. Let x_{jm} be the value for CMP j and metric m , and let $x_m^{\min}$ and $x_m^{\max}$ be the minimum and maximum across the included CMPs. The directionally normalized preference score p_{jm} is

$$p_{jm} = \begin{cases} \frac{x_{jm} - x_m^{\min}}{x_m^{\max} - x_m^{\min}}, & \text{if higher values are preferred,} \\ \frac{x_m^{\max} - x_{jm}}{x_m^{\max} - x_m^{\min}}, & \text{if lower values are preferred.} \end{cases}$$

Thus $p_{jm} = 0$ is the least preferred observed value and $p_{jm} = 1$ is the most preferred observed value for that metric and comparison set. It is a relative-preference score, not an acceptability

score. If all CMPs have the same value for a metric, they are tied at $p_{jm} = 1$. With inclusion indicator I_m and non-negative weight w_m , the aggregate score is

$$S_j = 100 \frac{\sum_m I_m w_m p_{jm}}{\sum_m I_m w_m}.$$

The companion Slick Spider (radar) view applies the same higher-is-better orientation to the six included scorecard metrics and repeats the 0–1 normalization within each selected OM. It is useful for seeing how the shape of CMP trade-offs changes among OMs, but radar shapes from different OMs do not show absolute changes in performance because each OM is scaled separately.

The contribution of metric m to the total is $C_{jm} = 100 I_m w_m p_{jm} / \sum_m I_m w_m$, so $S_j = \sum_m C_{jm}$. The [R implementation](#) reads the editable Figure 7 input, checks that every CMP has one value for every included metric, and writes both the totals and metric-level contributions.

The worked results in Table 5 include the established six scorecard metrics from Figure 7; the four added diagnostic columns are visible inputs with `include = FALSE` until their inclusion and weights are reviewed. The equal-weight view assigns one-sixth to every included metric. The balanced view assigns half the total weight to fishing performance—25% each to Catch and IACC—and half to stock condition—12.5% each to SB/SB_{MSY} , F/F_{MSY} , VB/VB_{2025} , and VB/VB_{MSY} . All inputs use the reference OM and the 2041–2050 summary window used in Figure 7.

Table 5: Equal-weight and balanced scorecard results for the eight performance-metric-quilt CMPs under the reference OM, 2041–2050. The balanced score assigns 25% each to Catch and IACC and 12.5% each to SB/SB_{MSY} , F/F_{MSY} , $VB/VB[2025]$, and $VB/VB[MSY]$.

Equal-weight rank	CMP	Equal-weight score	Balanced score	Balanced rank
1	HS+20 (MP29)	73.74	69.90	1
2	HSsym (MP45)	63.74	63.91	2
3	PR-20 (MP44)	52.56	52.42	3
4	PRsym (MP46)	48.42	46.88	5
5	HS-30 (MP47)	44.28	48.43	4
6	HS-20 (MP43)	27.66	36.78	6
7	PR+20 (MP32)	22.25	21.11	7
8	PR-30 (MP48)	16.85	12.64	8

The machine-readable totals are in [candidate_scorecard_result_reference.csv](#), and the individual weighted contributions are in [candidate_scorecard_contributions_reference.csv](#). These values are a reproducible calculation, but both orders remain **illustrative relative trade-offs rather than management rankings or acceptability scores**: neither weighting

scheme is an agreed statement of management priorities, and the scores are relative to this candidate set.

The current worked scorecard based on Figure 7 is an **illustrative framework**, not the final scorecard. It contains eight CMPs and ten displayed performance metrics, six of which are included in the worked composite scores. Please note:

- candidate rows can be added, removed, or replaced as the shortlist evolves;
- performance columns can be added as further safety, yield, stability, feasibility, and robustness metrics are validated;
- preferred direction, units, periods, transformation, and weight must be recorded for every metric; and
- normalization must be recalculated whenever the candidate set changes.

Min–max normalization is relative to the alternatives included. Adding or removing a CMP can therefore change every normalized score even when underlying MSE results do not change. This sensitivity is especially important here: because all eight CMPs have been tuned close to the same 60% dynamic Kobe-green objective, a narrow absolute range can still be stretched across the full 0–1 relative-preference scale. Absolute distances from agreed targets and the uncertainty in the raw outcomes are therefore more informative than min–max ordering when candidates are closely clustered. Weights express management priorities and should be agreed and recorded rather than presented as scientific constants. A weighted score is a decision aid, not a recommendation, and should always be shown with the Layer 1 underlying values and uncertainty, any defensible Layer 2 absolute scores, and sensitivity analyses.

Main performance and robustness findings

Reference comparison

The illustrative reference-OM scorecard in Table 5 provides the current eight-CMP comparison. HS+20 ranks first and HSsym second under both the equal-weight and balanced schemes. PR-20 ranks third under both. HS-30 and PRsym exchange fourth and fifth positions between the schemes, followed by HS-20, PR+20, and PR-30. The separation between the two **+20** sensitivity CMPs shows why an analytical role should not be confused with a shared performance ranking.

This ordering is illustrative and relative to the CMPs, metrics, directions, 2041–2050 window, and weights included in the scorecard. It does not mean that one CMP dominates every biological, yield, stability, and responsiveness objective. The underlying values and trade-off figures should therefore remain the basis for interpreting why the order changes with management priorities.

Reference set trade-offs

Kobe status. Figure 8 provides the direct reference-OM comparison of spawning-biomass status and fishing pressure for the near-term (2026–2035) and long-term (2041–2050) periods. Within each posterior iteration, each ratio is averaged over the stated period; points are the medians of those period means across the 500 iterations. The vertical and horizontal lines at 1 define the conventional Kobe quadrants.

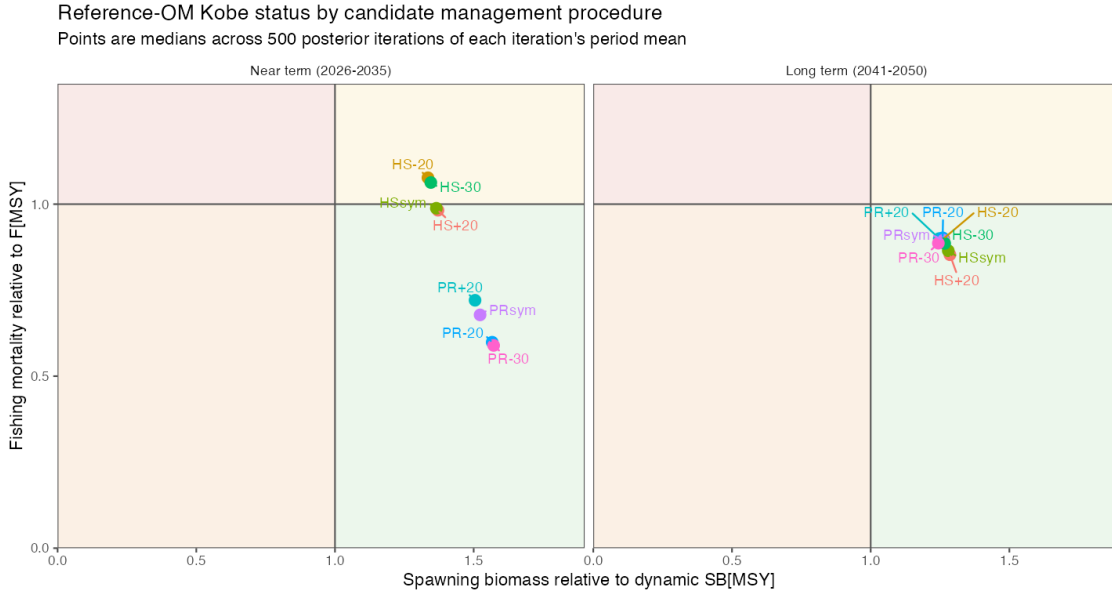


Figure 8: Near- and long-term Kobe plots for the eight CMPs under the reference operating model. The horizontal axis uses the time-varying dynamic SBMSY denominator used in tuning.

Vulnerable biomass. The near-term reference comparison uses mean annual catch and mean VB/VB_{MSY} over 2026–2035 (Figure 9). Points are CMP medians and bars are marginal 5th–95th percentiles.

The long-term reference comparison uses mean annual catch and mean VB/VB_{2025} over 2041–2050 with the same aggregation (Figure 10).

Spawning biomass. The reference-set yield–spawning-biomass comparison uses mean catch during 2026–2030 and mean SB/SB_{MSY} during 2036–2040. Points show means among 500 draws, and horizontal and vertical bars show the separate 10th–90th percentile ranges for the two measures.

Figure 11 shows the reference-OM pattern. The hockey-stick CMPs generally occupy the higher-catch, lower-biomass part of the plot, while the power-ramp CMPs generally produce lower catch and higher biomass. Responses to the annual-change variants are not monotonic

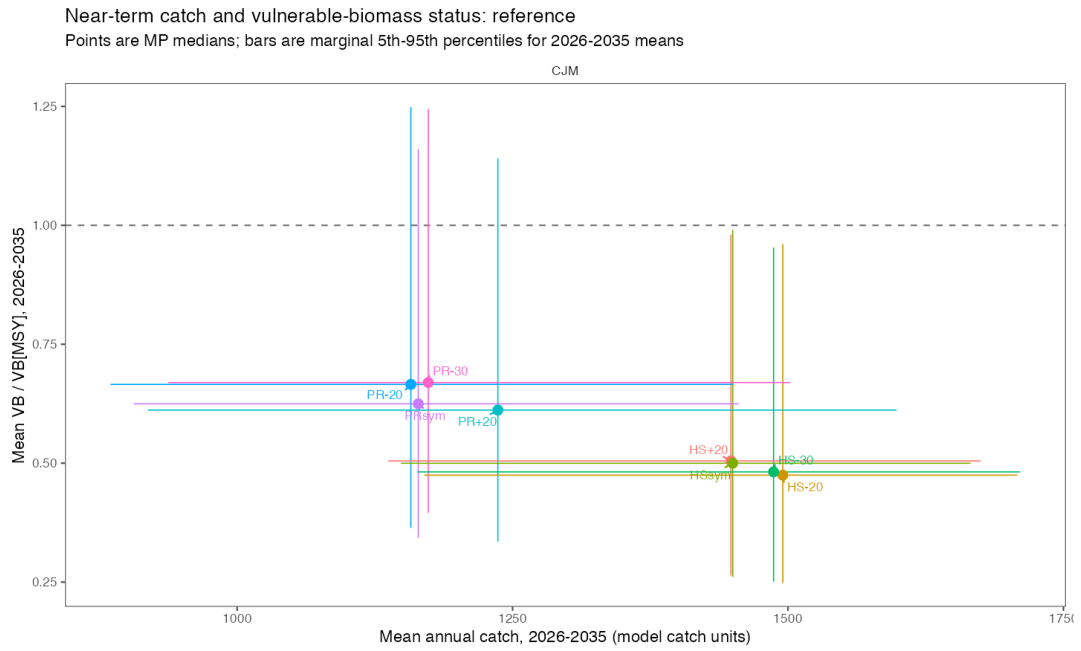


Figure 9: Near-term catch versus VB/VBMSY for the reference operating model, 2026–2035.

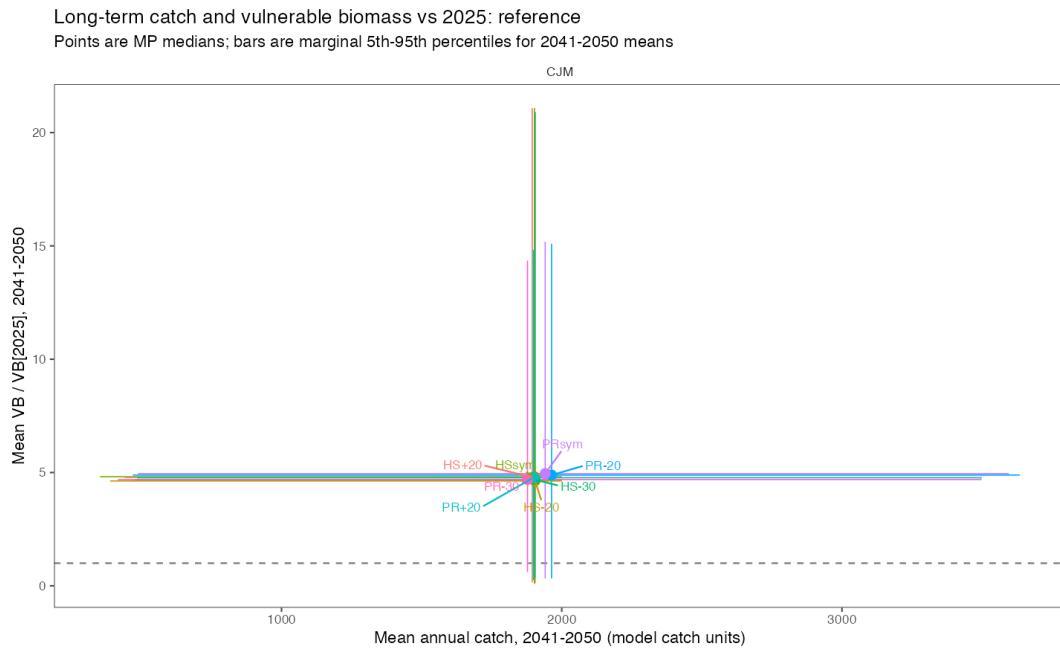
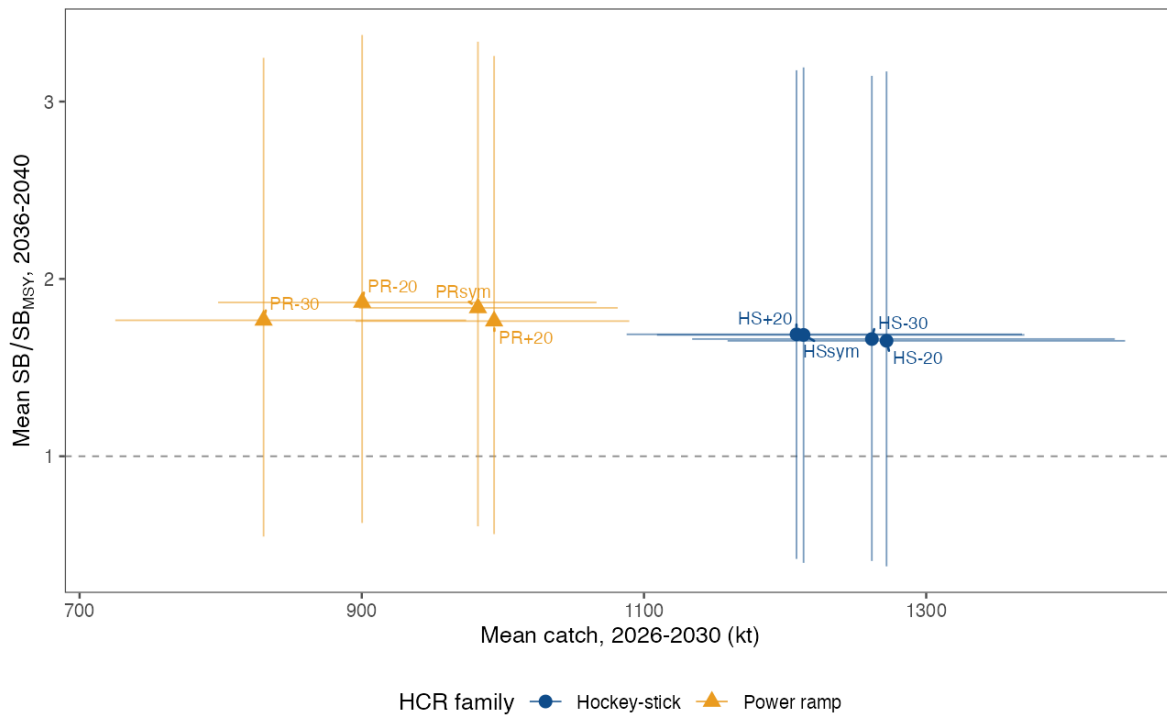


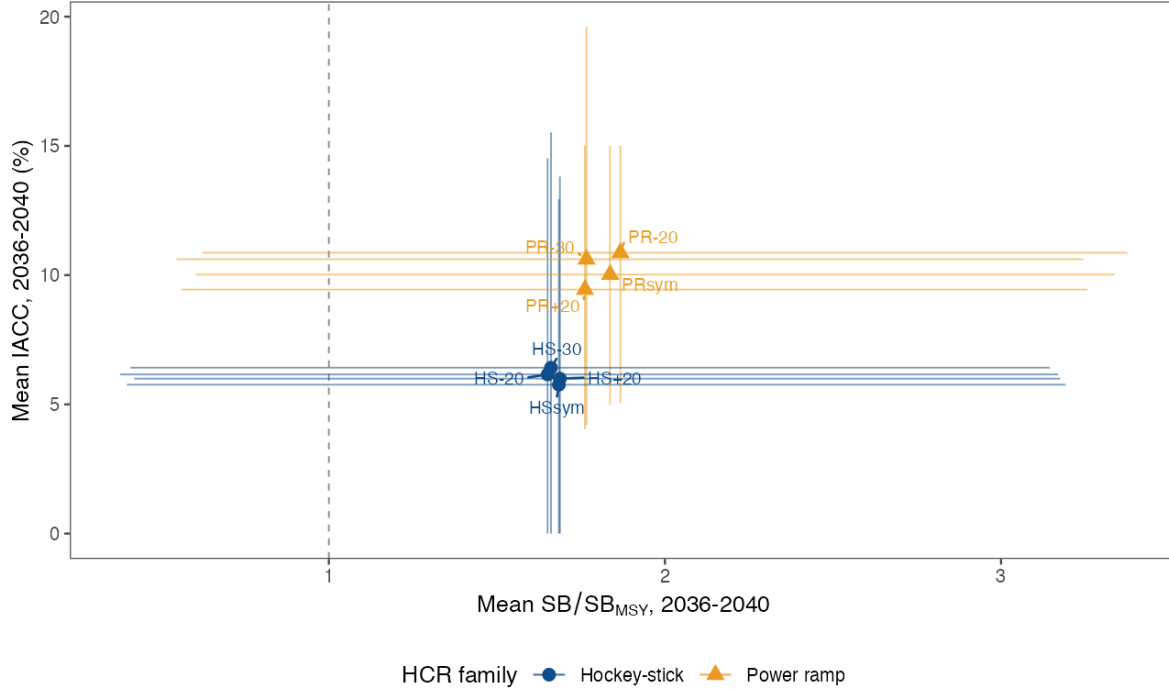
Figure 10: Long-term catch versus VB/VB2025 for the reference operating model, 2041–2050.



Points are means; bars show 10th-90th percentiles among 500 posterior draws. Dashed line: $SB/SB_{MSY} = 1$.

Figure 11: Reference-OM trade-off for the eight HS and PR configurations.

across the two families because each variant includes a separately tuned trigger as well as different change limits. The uncertainty ranges overlap substantially, so the plot should not be read as a deterministic ranking.



IACC is the absolute year-to-year percentage change in catch.
Points are means; bars show 10th-90th percentiles among 500 posterior draws. Dashed line: $SB/SB_{MSY} = 1$.

Figure 12: Reference-OM spawning-biomass and interannual-catch-variability trade-off for the eight HS and PR configurations.

Figure 12 provides a complementary reference-OM view of biological status and catch stability. The horizontal axis is the replicate mean SB/SB_{MSY} during 2036–2040. The vertical axis is mean IACC during the same period, where annual IACC is $100|C_t/C_{t-1} - 1|$: the plotted period therefore covers the five annual changes from 2035–2036 through 2039–2040. Lower IACC indicates more stable catch. Points are means among 500 draws; horizontal and vertical bars are the separate 10th–90th percentile ranges of the replicate-level period means. As in Figure 11, overlap is substantial and the marginal bars do not show within-replicate covariance.

Trade-offs under robustness cases

Recruitment stress

The discrete recruitment crash and seven-year cyclic reduction lowered long-term catch and increased catch variability for both central CMPs. PR+20 responded more quickly and produced

comparatively stronger long-term recovery in these tests, while HS+20 retained the relative stability advantages seen in the reference case. These are robustness contrasts, not forecasts that either recruitment path will occur.

Alternative selectivity

The alternative-selectivity OM exposed younger and smaller fish to greater fishing pressure. Similar catch levels could then occur with lower spawning biomass and poorer Kobe performance. In the completed results, HS+20 retained higher catch while PR+20 retained higher spawning biomass; this is a trade-off, not evidence that either CMP is uniformly better. Interpretation is also affected by the changed F_{MSY} reference point.

Two-stock hypotheses and catch distribution

In the two-stock OM without movement, the southern component remained above the agreed benchmark while the northern component did not. An exploratory shift of ~3% in simulated catch distribution from north to south brought both components to the 60% long-term benchmark. This illustrates sensitivity to spatial exploitation; it is not an allocation or catch-distribution recommendation.

In the movement OM, biomass transferred from the southern to the northern component. Because the five indices were combined with equal weights, increasing northern indices could raise the overall indicator while southern spawning biomass declined. No tested time-invariant catch distribution resolved the long-term benchmark for both components. The test is deliberately stressful and does not establish that this movement process is occurring. It does establish a management-relevant warning signal: persistent divergence between northern and southern indices.

Vulnerable biomass. The near-term catch and VB/VB_{MSY} comparison across robustness OMs is shown in Figure 13. It uses the same 2026–2035 period, CMP medians, and marginal 5th–95th percentiles as the reference comparison.

The long-term robustness comparison in Figure 14 uses mean annual catch and mean VB/VB_{2025} over 2041–2050.

Spawning biomass. The saved results also support yield–spawning-biomass comparisons across the single-stock and two-stock robustness cases.

The same broad direction is visible across the single-stock robustness cases (Figure 15), but the location and separation of configurations changes by OM. Recruitment stress reduces later biomass and changes the apparent trade-off, while productivity and assessment-configuration alternatives move both axes.

For the two-stock cases (Figure 16), each row is a biological component and the horizontal axis is **component catch**, not total catch. The plots therefore diagnose stock-specific consequences

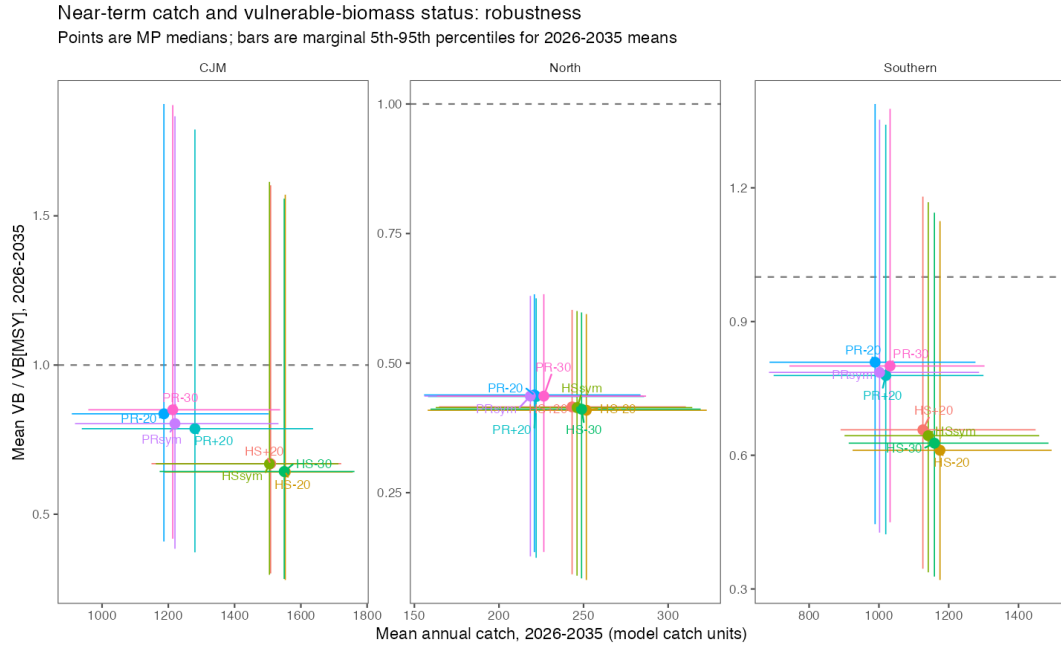


Figure 13: Near-term catch versus VB/VBMSY across robustness operating models, 2026–2035. CJM points pool the five single-stock robustness OMs; Northern and Southern points separately pool the corresponding components of the four two-stock OMs.

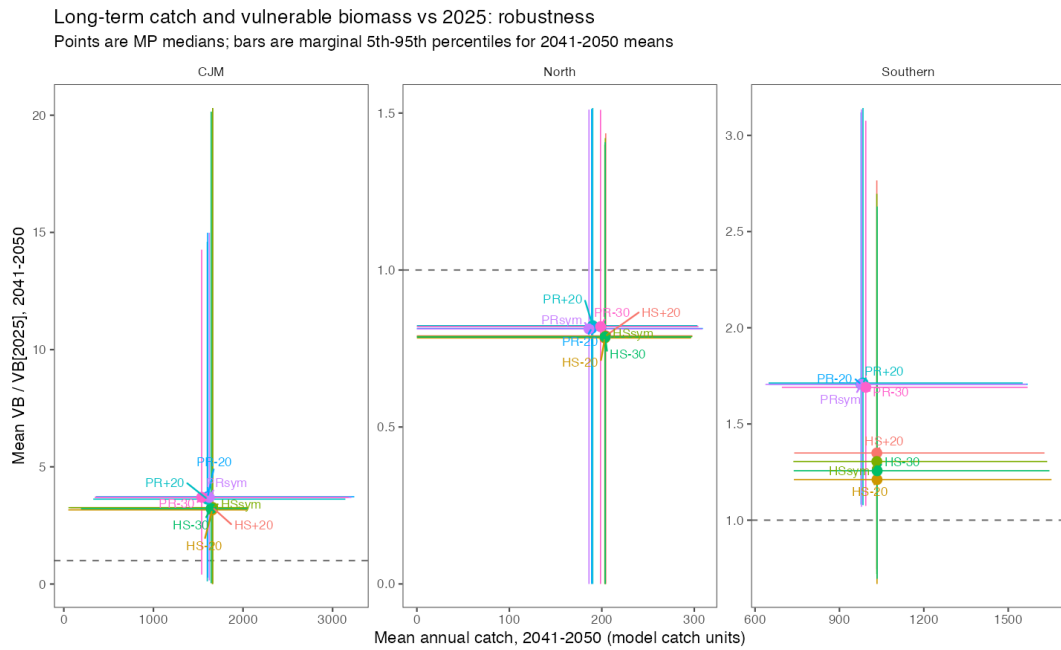
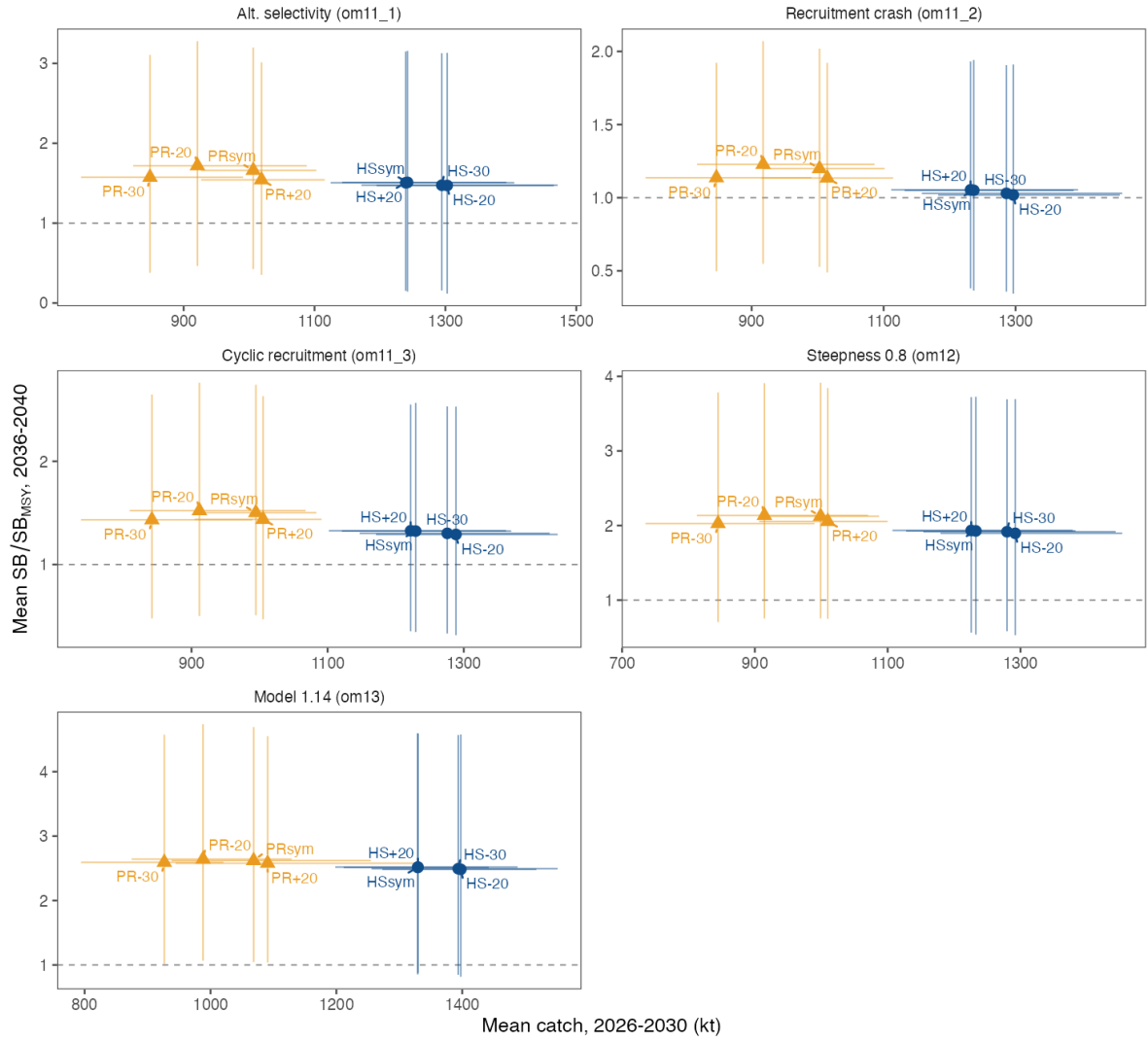


Figure 14: Long-term catch versus VB/VB2025 across robustness operating models, 2041–2050. CJM points pool the five single-stock robustness OMs; Northern and Southern points separately pool the corresponding components of the four two-stock OMs.



HCR family ● Hockey-stick ▲ Power ramp
 Points are means; bars show 10th-90th percentiles among 500 posterior draws. Dashed line: SB/SB_{MSY} = 1.

Figure 15: Trade-offs across the five single-stock robustness OMs.

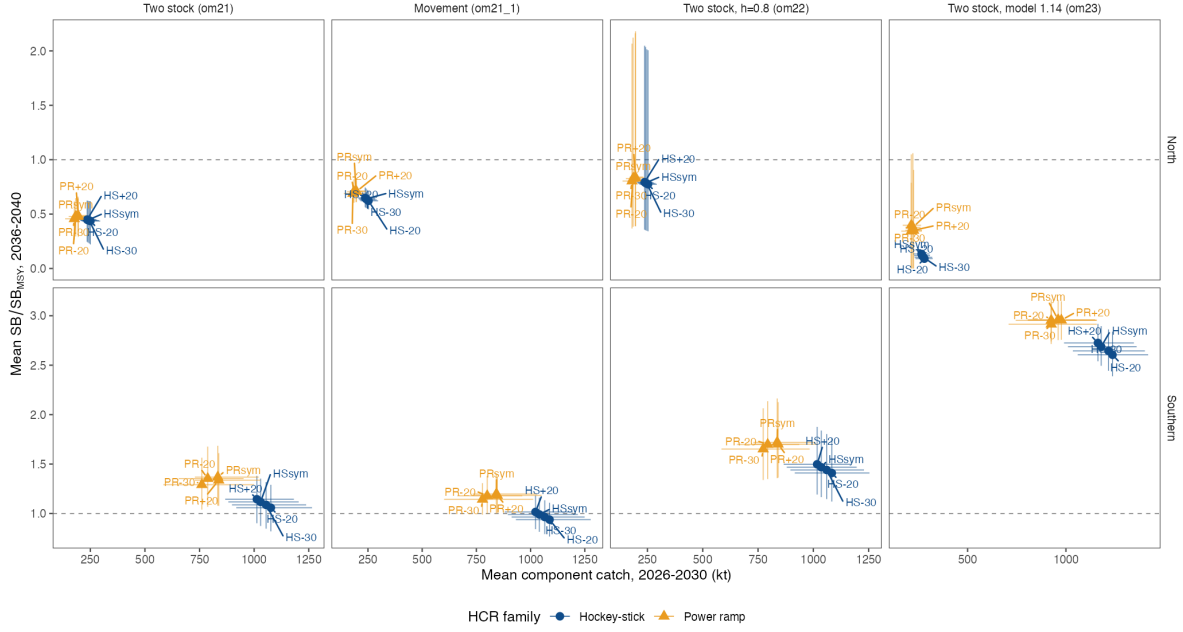


Figure 16: Component-level trade-offs across the four two-stock robustness OMs.

and must not be interpreted as an allocation comparison. They reinforce the need to keep northern and southern outcomes separate.

The spawning-biomass figures in Sections 6.2 and 6.3 use the completed 500-draw comparison but remain selective summaries for three reasons. First, the x and y uncertainty bars are marginal percentiles and do not show the within-replicate covariance between plotted measures. Second, spawning biomass is shown relative to each draw's equilibrium SBMSY rather than in tonnes. This differs from the time-varying dynamic SBMSY denominator used for tuning and in Figure 8. Third, these views cover only selected yield, biological, and catch-stability periods; they do not include fishing pressure, Kobe-green probability, low-catch risk, feasibility, or the other performance measures needed for a decision. The plotted summary values are saved in `doc/data/sc14-preliminary-tradeoffs.csv` and can be regenerated with `R/plot_sc14_tradeoffs.R`.

Banking, borrowing, and implementation constraints

SCW17 treated banking and borrowing as a sensitivity to apply to the final CMP set rather than as feedback within the HCR. Preliminary 100-iteration results are now available for HS+20 and PR+20 under the reference, low-recruitment, and cyclic-recruitment OMs. The tested implementation used a 10% banking or borrowing amount when unconstrained HCR advice changed by at least 15%, subject to the CMP-specific healthy-rule condition. In operational

terms, a decline of more than 15% made an iteration eligible to **borrow** 10% of the new advice (adding it to the current TAC and deducting it subsequently), while an increase of more than 15% made it eligible to **bank** 10% (deducting it from the current TAC and making it available subsequently). Eligibility was restricted to years in which the combined abundance indicator was in the upper, or “healthy’’, branch of the applicable HCR. For HS+20 this is rule 3: the indicator is at or above the hockey-stick trigger and the unconstrained advice is on the constant target branch. For PR+20 this is rule 2: the indicator is at or above the power-ramp trigger and advice is on its upper branch. Thus, a change exceeding 15% was necessary but was not by itself sufficient. These runs did **not** retain the annual TAC-change limits, so they do not isolate banking and borrowing from the effect of removing those limits.

The contractor-reported long-term results are summarized in Table 6. Under the reference OM, mean SB0green during 2041–2050 declined from 0.603 to 0.589 for HS+20 and from 0.607 to 0.585 for PR+20. The reported banking-and-borrowing values under low recruitment were 0.530 and 0.617, respectively, and under cyclic recruitment were 0.419 and 0.539.

Table 6: Contractor-reported mean SB0green during 2041–2050 for the preliminary 100-iteration banking-and-borrowing sensitivity. A dash indicates that the corresponding no-B&B comparison was not supplied in the contractor summary.

OM	CMP	Without B&B	With B&B
Reference OM	HS+20	0.603	0.589
Reference OM	PR+20	0.607	0.585
Low recruitment	HS+20	–	0.530
Low recruitment	PR+20	–	0.617
Cyclic recruitment	HS+20	–	0.419
Cyclic recruitment	PR+20	–	0.539

The spawning-biomass distributions for the six banking-and-borrowing runs in Table 6 are shown in Figure 17. Each boxplot summarizes the distribution across 100 iterations after first averaging SB/SBMSY within each iteration over the stated period. This is a stock-status ratio, whereas SB0green in Table 6 is the probability of being in the dynamic Kobe green quadrant; the two summaries are complementary and should not be read as the same performance measure. The dashed line marks $SB/SBMSY = 1$.

The transferred run archive requires an implementation check before these values are treated as final. Recalculation from the supplied `babs.rds` gives some different SB0green values, and its saved `banking.isys` and `borrowing.isys` tracking fields contain no non-zero transactions. Inspection of the archived implementation shows that the intended previous HCR advice is retrieved for the current year, which makes the transaction test compare the current value with itself. Figure 18 therefore shows **reconstructed eligibility events**, based on the intended comparison with the preceding year’s HCR advice; it must not be interpreted as a count of transactions actually applied in the archived runs.

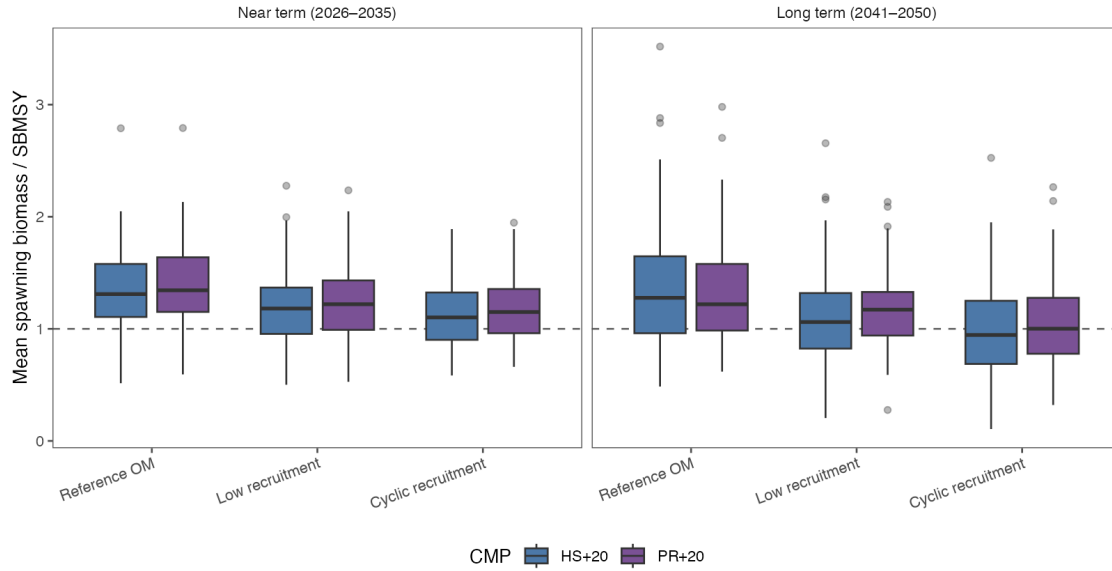


Figure 17: Spawning biomass relative to SBMSY in the preliminary banking-and-borrowing runs, for the near-term (2026–2035) and long-term (2041–2050) periods.

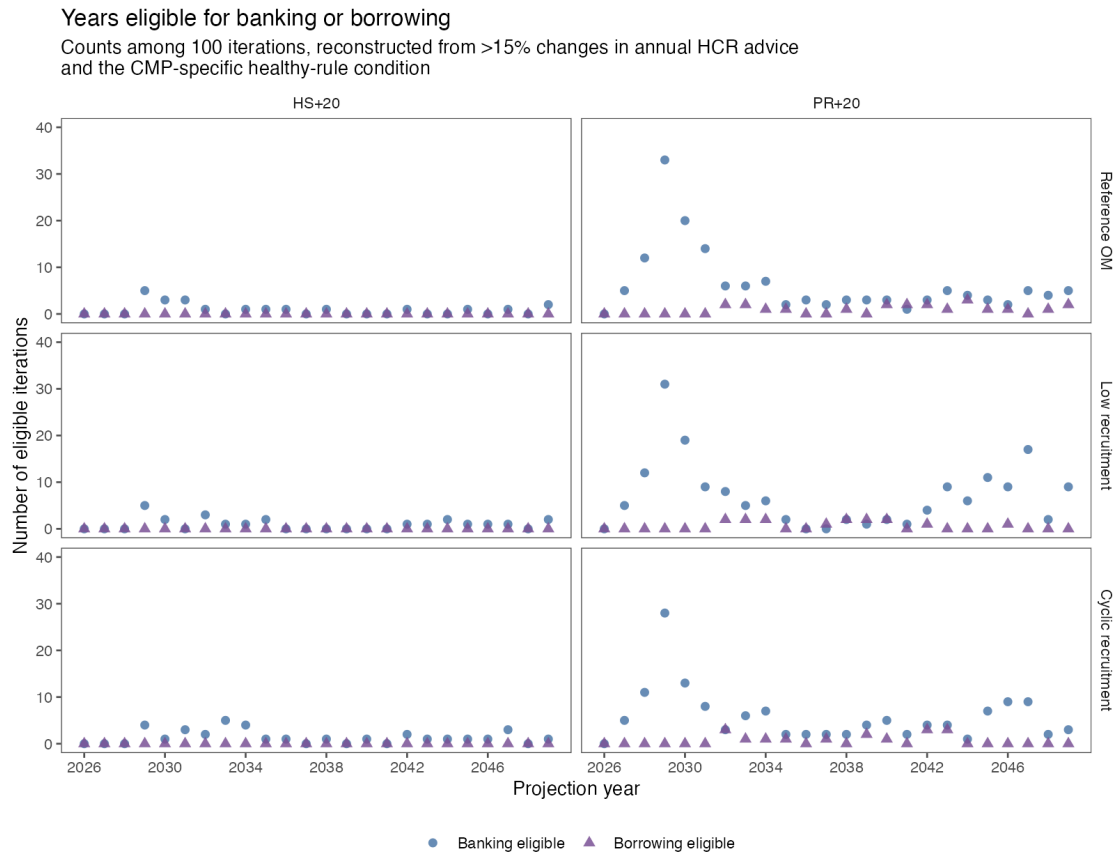


Figure 18: Reconstructed annual eligibility for banking and borrowing in the preliminary 100-iteration runs. Circles denote iterations eligible for banking and triangles denote iterations eligible for borrowing.

Final advice should use a corrected rerun that retains explicit transaction tracking and, if the intent is to estimate the incremental effect of banking and borrowing, includes a matched comparison with the same TAC-change-limit treatment.

Effort feasibility is another implementation constraint. A TAC can be mathematically available but difficult to catch when fish distribution or catchability changes. The OM includes an effort cap to prevent unrealistic projected effort. Low realized catch, catch-rate proxies, and the frequency of binding effort constraints should be retained as supporting diagnostics.

Exceptional circumstances and annual monitoring

An exceptional-circumstances protocol should identify when new information is inconsistent with the conditions under which a selected CMP performed acceptably. SCW17 considered the ECP software framework as a proof of concept: robustness simulations can identify data patterns associated with poor CMP performance, and detection rules can then be evaluated for false-positive and false-negative errors.

The current analysis identifies at least two monitoring priorities:

1. persistent divergence between northern and southern abundance indicators, particularly in relation to the two-stock-with-movement hypothesis; and
2. whether current survey, catch, and fishery information lies outside OM distributions or is associated with simulated CMP failure states.

Preliminary oral fishery updates reported at the 24 July meeting are not validated data submissions and do not by themselves establish exceptional circumstances. A protocol should specify the observations, data timing, triggering evidence, review authority, possible responses, and return-to-rule conditions before operational use.

Summary and conclusions

This MSE combines the operating-model development and management-procedure design undertaken before and during SCW17 with the subsequent implementation, tuning, robustness testing, and performance evaluation. The candidate procedures apply transparent hockey-stick and power-ramp HCRs to a combined indicator constructed from five standardized abundance series. A distinctive feature is that the OMs simulate uncertainty, temporal autocorrelation, and cross-correlation among those series, rather than treating multiple indicators as independent confirmations of stock change.

The analysis narrows the broader workshop set to two central rule families and reasonable annual TAC-change variants. Candidate performance is evaluated against a reference OM and a structured robustness set spanning productivity, recruitment stress, selectivity, stock structure, movement, and spatial catch distribution. The results demonstrate trade-offs among biological

condition, catch, stability, and responsiveness rather than identifying a single CMP that dominates every objective. The performance quilt and worked scorecards provide transparent summaries, but the underlying distributions, metric definitions, weights, and robustness results remain necessary for interpretation. For example, MP29 and MP45 (HS+20 and HSSym) receive the two highest illustrative composite scores under the reference OM. That result is not a claim that either dominates across the robustness set: their relative performance still depends on the metric, weighting, and robustness scenario considered.

The [external experts invited to SCW17](#) supported progressing the MSE. Tom Carruthers considered the framework suitable for testing CMPs, with a defensible reference OM, a comprehensive robustness set, relevant performance measures, and empirical CMPs comparable with procedures used in other fisheries. He concluded that the work was sufficiently advanced to support shortlisting defensible CMPs for Scientific Committee consideration and that the improvements still identified were not substantial enough to delay adoption of a robust management procedure. He also emphasized that MSE is an ongoing, iterative process in which the framework can be refined after an MP is adopted and applied.

David Miller and Rosana Ourens likewise viewed the benchmark assessment as a strong basis for the reference OM, the robustness set as covering important uncertainties, and the use of dynamic reference points and simple empirical index-based rules as sensible. Their comments were made at a particular stage of the workshop process and should not be read as a complete account of stakeholder engagement. Commission participants and other stakeholders had received several presentations and opportunities to provide input before and during the MSE process, including input on management preferences and annual TAC-change limits. Suggestions for further engagement are therefore best understood as support for continuing and documenting that involvement, not as an indication that stakeholder input was absent.

The technically relevant priorities identified by the experts include accurately representing historical cross-correlation among indices, exploring alternative index combinations, and establishing a formal exceptional-circumstances framework. Taken together, the external review supports applying the MSE as a living management process, with monitoring, stakeholder input, and agreed revisions continuing after initial adoption rather than treating the current analysis as a one-time exercise.

The available work supports taking a reduced subset of CMPs forward for Commission selection, accompanied by annual monitoring and a formal exceptional-circumstances process. Because this synthesis contains result-dependent statements, scores, and conclusions, **this section must be updated dynamically in future report versions whenever simulation results, candidate definitions, performance metrics, or agreed weights change.**

The Scientific Committee is asked to consider the following:

Noting

- that the approaches provide valid combinations of the available information for application in the tested HCRs;

- that the work identifies reasonable annual TAC-change variants informed by polling of Commission members;
- that the results satisfy the Commission requests and recorded management preferences;
- the use of all five indices and the complementary value of the fishery-independent acoustic series;
- that northern and southern indicators should continue to be tracked separately, as in the past, alongside the combined CMP indicator and monitored through the annual assessment process, including when evaluating whether exceptional circumstances may be occurring;
- that the preliminary banking-and-borrowing results suggest effects of a manageable magnitude at the investigated levels, but require a corrected, matched rerun before implementation conclusions are drawn; and
- that the robustness tests provide plausible scenarios for testing the CMPs.

Endorsing

- the analytical work as satisfying the request from the Commission;
- adoption of a subset of the CMPs presented so that the Commission may select one among them to satisfy the trade-offs it may consider;
- development of a formal exceptional-circumstances protocol, including evaluation of detection performance and pre-agreed responses; and
- review and documentation of the full performance-metric set and any weighting process, while retaining the underlying outcomes rather than relying on a composite score alone.

Work required before final advice

i Note

The remaining analytical and documentation tasks are summarized in Table 7.

Table 7: Work required before final scientific advice.

Work item	Why it matters	Current status in source material
Archive and independently reproduce the completed 500-draw tuning traces	Confirms the common tuning objective and preserves an auditable record of the refined controls	Eight CMPs refined under one frozen OM and common random-number configuration; archive review remains

Review the TAC-change comparison and document the shortlist	Connects the completed responsiveness and stability results to the alternatives advanced for Commission consideration	Common 500-draw reference and nine-OM robustness results available; interpretation and shortlist decision remain
Validate and complete banking-and-borrowing sensitivity	Tests an implementation provision outside the HCR	Preliminary contractor results available; transferred archive records no transactions and requires a corrected, matched rerun
Add index-error parameter distributions	Makes observation uncertainty auditable	Requested for technical annex
Improve index and HCR diagnostics	Clarifies responsiveness and the acoustic survey's role	Requested for technical annex
Finalize metric definitions, periods, and absolute targets where defensible	Prevents ambiguous or directionally incorrect scoring	Three-layer framework documented; management targets and tolerances remain to be agreed
Sensitivity-test the scorecard	Shows how weights, candidate membership, and relative scaling affect the illustrative ordering	Reproducible relative-preference calculation available; sensitivity analysis remains
Specify exceptional-circumstances protocol	Defines detection, authority, response, and return to the CMP	Proof-of-concept stage
Freeze reproducible inputs and outputs	Supports independent review and exact reruns	Run records exist; final archive still required

Source, terminology, and interpretation notes

This synthesis uses “CMP” for a procedure still under evaluation and “MP” when referring to the identifiers used in the analysis. “Catch distribution” is used for two-stock sensitivity tests; those tests are not allocation advice. “Dynamic Kobe green” refers to performance calculated using the dynamic reference-point implementation described above.

Numerical values from presentations and meeting materials should be checked against the frozen final simulation archive before submission. Where a presentation takeaway and the later meeting interpretation differed in tone, this report follows the meeting's more cautious interpretation. The meeting report itself was prepared from meeting notes, transcript material, and an automated meeting summary; preliminary oral information requires confirmation against formal submissions.

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